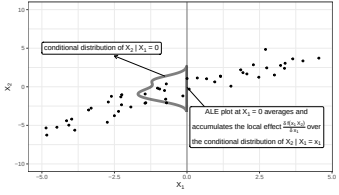
Interpretable Machine Learning

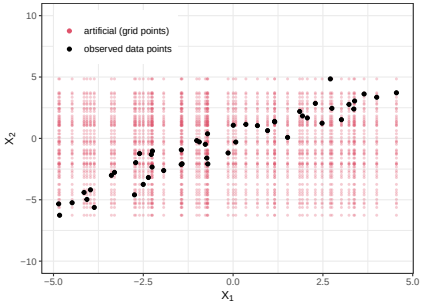
Accumulated Local Effect (ALE): Introduction

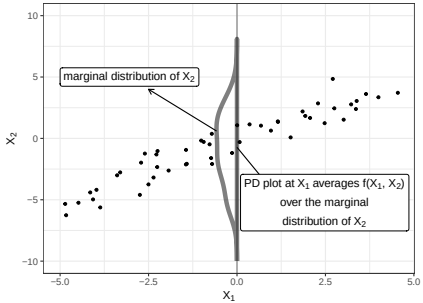


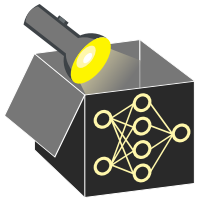
Learning goals

- PD plots and its extrapolation issue
- M plots and its omitted-variable bias
- Understand ALE plots

MOTIVATION - CORRELATED FEATURES

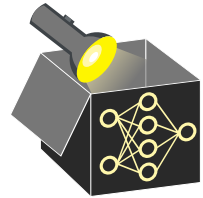






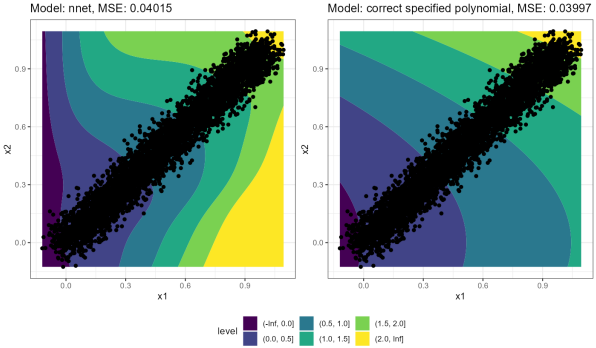
- PD plots **average over predictions** of artificial points that are out of distribution/ unlikely (red)
  $\Rightarrow$ Can lead to misleading / biased interpretations, especially if model also contains interactions
- Not wanted if interest is to interpret effects within data distribution

MOTIVATION - CORRELATED FEATURES



Example: Fit an NN to 5000 simulated data points with $x \sim Unif(0, 1)$, $\epsilon \sim N(0, 0.2)$ and

$$y = x_1 + x_2^2 + \epsilon, \text{ where } x_1 = x + \epsilon_1,\ x_2 = x + \epsilon_2 \text{ and } \epsilon_1, \epsilon_2 \sim N(0, 0.05).$$



- Test error (MSE) of NN is comparable to other models
- NN contains interactions (see complex pred. surface)

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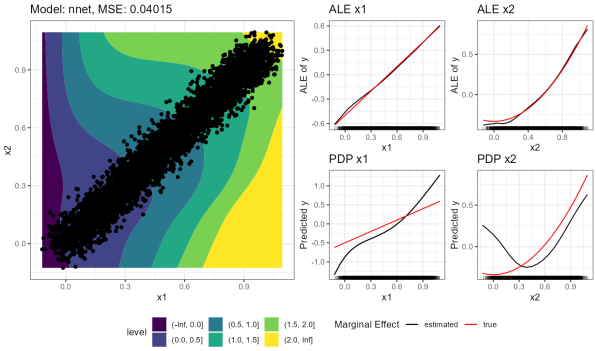
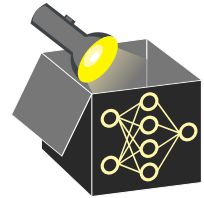
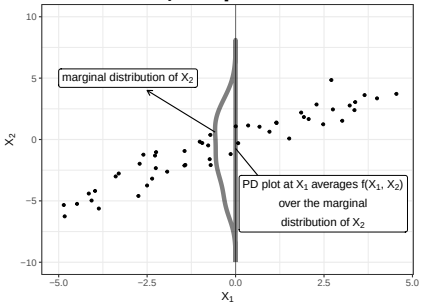


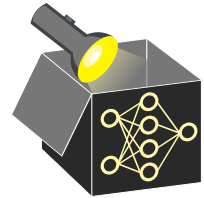


- Test error (MSE) of NN is comparable to other models
- NN contains interactions (see complex pred. surface)
- ALE in line with ground truth
- PDP does not reflect ground truth effects of DGP well
  $\Rightarrow$ Due to interactions and averaging of points outside data distribution

M PLOT VS. PD PLOT

a) PD plot





a) PD plot $\mathbb{E}_{\mathbf{x}_2}\left(\hat{f}(x_1, \mathbf{x}_2)\right)$ is estimated by $\hat{f}_{1,PD}(x_1) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}(x_1, \mathbf{x}_2^{(i)})$

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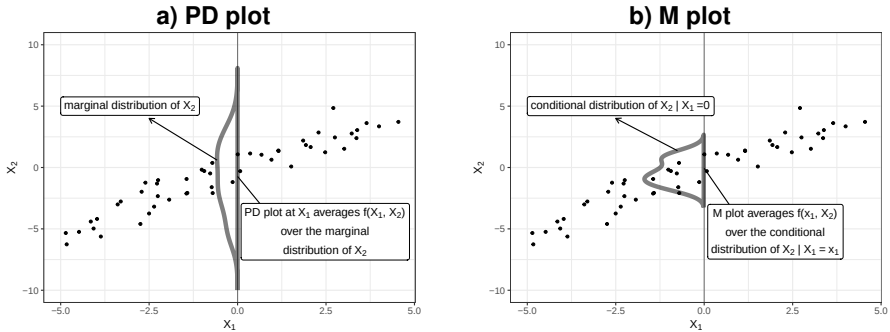

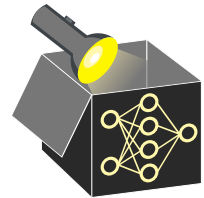

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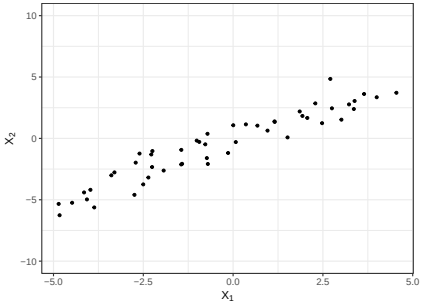
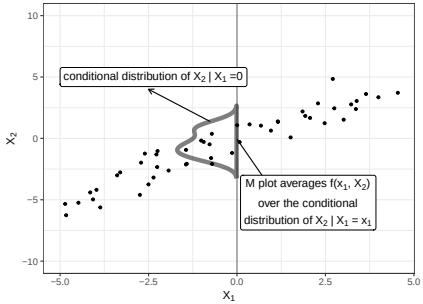
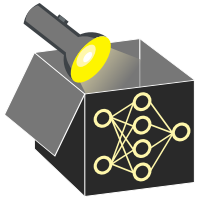
b) M plot $\mathbb{E}_{\mathbf{x}_2|\mathbf{x}_1}\left(\hat{f}(x_1, \mathbf{x}_2)\middle|\mathbf{x}_1\right)$ is estimated by $\hat{f}_{1,M}(x_1) = \frac{1}{|N(x_1)|}\sum_{i \in N(x_1)} \hat{f}(x_1, \mathbf{x}_2^{(i)})$, where index set $N(x_1) = \{i : x_1^{(i)} \in [x_1 - \epsilon, x_1 + \epsilon]\}$ refers to observations with feature value close to x_1.

M PLOT VS. PD PLOT







- M plots average predictions over conditional distribution (e.g., $\mathbb{P}(\mathbf{x}_2|x_1)$)
  $\Rightarrow$ Averaging predictions close to data distribution avoid extrapolation issues
- **But:** M plots suffer from omitted-variable bias (OVB)
  - Because of the conditioning M plots contain effects of other dependent features
  - Useless in assessing a feature's marginal effect if feature dependencies are present

M PLOT VS. PD PLOT - OVB EXAMPLE

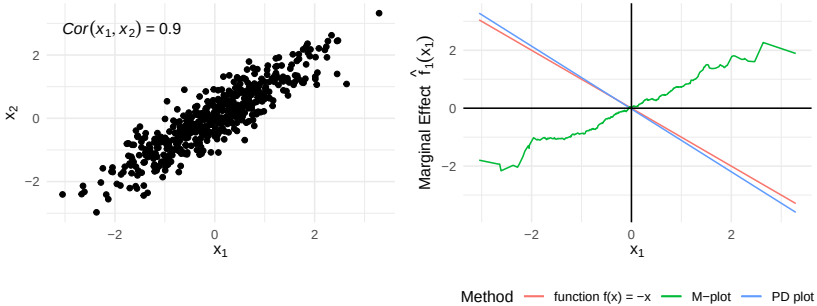
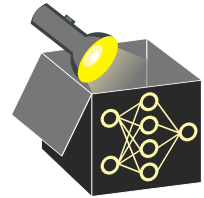


Illustration: Fit LM on 500 i.i.d. observations with features $x_1, x_2 \sim N(0, 1)$, $Cor(x_1, x_2) = 0.9$ and

$$y = -x_1 + 2 \cdot x_2 + \epsilon, \ \epsilon \sim N(0, 1).$$

Results: M plot of x_1 also includes marginal effect of all other dependent features (here: x_2)

IDEA: INTEGRATING PARTIAL DERIVATIVES



Idea: To remove unwanted effects of other features, take partial derivatives (local effects) of prediction function w.r.t. feature of interest and integrate (accumulate) them w.r.t. the same feature

$\Rightarrow$ Computing the partial derivative of $\hat{f}$ w.r.t. $\mathbf{x}_j$ removes other main effects

$\Rightarrow$ Integrating again w.r.t. $\mathbf{x}_j$ recovers the original main effect of $\mathbf{x}_j$

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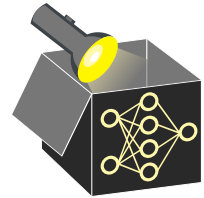

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$$\hat{f}(x_1, x_2) = 2x_1 + 2x_2 - 4x_1x_2$$

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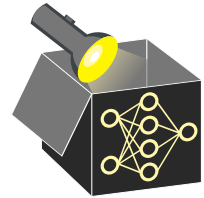

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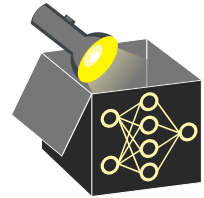

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$$\int_{z_0}^{x} \frac{\partial \hat{f}(x_1, x_2)}{\partial x_1} dx_1 = [2x_1 - 4x_1x_2]_{z_0}^{x}$$

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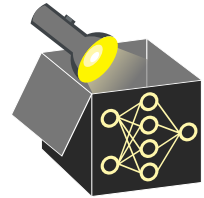

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- We removed the main effect of x_2, which was our goal